Name: _____.

Student number: _____.

**Instructions**

- Turn off all the electronic devices.
- This is a closed book, and no notes exam.
- The last page of this exam contains a notesheet that you can tear off.
- You can use any fact in the notesheet, but you must reference which fact you are using.
- You can use every theorem proved in class, but you need to state the theorem.
- Unless otherwise stated, you must justify your answers.
- If you have a question, raise your hand and I will come to you.
- You have 1 hour and 50 minutes to complete the exam.
- Good luck!

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Points: | 10 | 10 | 10 | 10 | 10 | 10 | 60 |
| Score: | | | | | | | |

1. (10 points) Let $r \in \mathbb{R}$ with $r \neq 1$. Prove that $\sum_{i=0}^{n} r^i = \frac{1-r^{n+1}}{1-r}$ for every integer $n \geqslant 0$.

Proof: By induction on $n \geq 0$:

Base case $n=0$: $\sum_{i=0}^{n} r^i = \sum_{i=0}^{0} r^i = r^0 = 1$

and $\frac{1-r^{n+1}}{1-r} = \frac{1-r}{1-r} = 1$, so base case holds.

Assume that $\boxed{\sum_{i=0}^{k} r^i = \frac{1-r^{k+1}}{1-r}}$ for some $k \geq 0$. Ind Hyp

Now, to show the statement is true for $k+1$, we compute

$$\sum_{i=0}^{k+1} r^i = \sum_{i=0}^{k} r^i + r^{k+1}$$

$$= \frac{1-r^{k+1}}{1-r} + r^{k+1} \quad \text{(Ind Hyp)}$$

$$= \frac{1-r^{k+1}+(1-r)r^{k+1}}{1-r}$$

$$= \frac{1-r^{k+1}+r^{k+1}-r^{k+2}}{1-r}$$

$$= \frac{1-r^{k+2}}{1-r}, \text{ as desired.}$$

So, by induction, the statement holds for every $n \geq 0$ □

2. (10 points) Consider the function $f: \mathbb{Z} \to \mathbb{Z}$ defined by

$$f(x) = \begin{cases} x+2 & \text{if 3 divides } x \\ x-1 & \text{otherwise} \end{cases}$$

(a) Prove that f is injective. Let $x_1, x_2 \in \mathbb{Z}$ such that $f(x_1) = f(x_2)$. We want to show that $x_1 = x_2$. We consider 3 cases:

Case 1: Both $x_1$ and $x_2$ are divisible by 3. Then $f(x_1) = x_1 + 2$ and $f(x_2) = x_2 + 2$. Since $f(x_1) = f(x_2)$, we have $x_1 + 2 = x_2 + 2$, hence $x_1 = x_2$, as desired.

Case 2: If neither $x_1$ nor $x_2$ are divisible by 3, Then $f(x_1) = x_1 - 1$ and $f(x_2) = x_2 - 1$. Now, since $f(x_1) = f(x_2)$ we have $x_1 - 1 = x_2 - 1$, hence $x_1 = x_2$

Case 3: If only one of them is divisible by 3, say $3 \mid x_1$ but $3 \nmid x_2$, Then $f(x_1) = x_1 + 2$ and $f(x_2) = x_2 - 1$. Hence $x_1 + 2 = x_2 - 1$, which means $x_1 - 3 = x_2$. But Then $3 \mid x_2$, which is a contradiction. Hence case 3 never occurs

(b) Prove that f is surjective.

Let $y \in \mathbb{Z}$. We want to find $x \in \mathbb{Z}$ such that $f(x) = \mathbb{Z}$.

We consider three cases for $y$; which cover all possibilities by Div. Thm

Case 1: $y \equiv 0 \pmod 3$. Then let $x = y + 1$. Since $x \equiv 1 \pmod 3$, we have $3 \nmid x$ so $f(x) = x - 1 = y$.

Case 2: $y \equiv 1 \pmod 3$. Again, let $x = y + 1$. Since $x \equiv 2 \pmod 3$, we have $3 \nmid x$, so $f(x) = x - 1 = y$.

Case 3: $y \equiv 2 \pmod 3$. Now let $x = y - 2$. In this case, $x \equiv 0 \pmod 3$ so $3 \mid x$, Thus $f(x) = x + 2 = y$

In any Case, we found $x \in \mathbb{Z}$ such that $f(x) = y$. Since $y \in \mathbb{Z}$ is arbitrary, f is surjective.

3. (10 points) Determine whether the following statements are true or false. If the statement is true, provide a brief proof. If the statement is false, provide a counterexample

(a) If A and B are sets, then $A \setminus B$ and $B \setminus A$ are disjoint.

This is True. Suppose for a contradiction that $(A \setminus B) \cap (B \setminus A) \neq \emptyset$

Then there exists some $x \in (A \setminus B) \cap (B \setminus A)$.

Since $x \in A \setminus B$, we have $x \in A$ but $x \notin B$.

Since $x \in B \setminus A$, we have in particular $x \in B$, which contradicts that $x \notin B$. Hence $(A \setminus B) \cap (B \setminus A) = \emptyset$. □

(b) Let $f: \mathbb{N} \to \mathbb{N}$ be a function. If f is injective, then it is surjective.

This statement is false, we provide a counterexample:

Let $f: \mathbb{N} \to \mathbb{N}$ given by $f(n) = 2n$.

Then f is injective: if $f(n) = f(m)$ then $2n = 2m$, and multiplying by $2^{-1} \in \mathbb{R}$ we get $n = m$.

But f is not surjective. In fact, since $1 \in \mathbb{N}$ is odd, we know it can't be even, so $1 \neq f(n)$ $\forall n \in \mathbb{N}$.

(c) Let $A = \{n \in \mathbb{Z} : n \equiv 3 \pmod 8\}$. If $x \in A$, then x is odd.

True statement. Let $x \in A$. Since $x \equiv 3 \pmod 8$,

we have $8 \mid (x-3)$, so there exists $k \in \mathbb{Z}$ such that $x - 3 = 8k$

Hence $x = 8k + 3 = 2(4k+1) + 1$ where $4k+1 \in \mathbb{Z}$. Hence x is odd.

4. (10 points) For each $m \in \mathbb{N}$, consider the set $A_m = \{mn \mid n \in \mathbb{N}\}$.

   (a) Is it true that $A_m \approx \mathbb{N}$ for all $m \in \mathbb{N}$? Either prove it or disprove it.

IT is True. Fix $m \in \mathbb{N}$. Define $f: \mathbb{N} \to A_m$ by $f(n) = mn$ for all $n \in \mathbb{N}$.
This $f$ is injective: if $f(n_1) = f(n_2)$ Then $mn_1 = mn_2$. Since $m \geq 1$, we know $m \neq 0$ so we have $m^{-1} \in \mathbb{R}$ and multiplying both sides of $mn_1 = mn_2$ by $m^{-1}$ we get $n_1 = n_2$. So $f$ is injective.
IT is also surjective: by definition of $A_m$, an arbitrary element of $A_m$ is the form $mn = f(n)$ for some $n \in \mathbb{N}$.
Hence $f$ is bijective and $\mathbb{N} \approx A_m$. So also $A_m \approx \mathbb{N}$.

   (b) Find $\bigcup_{m \in \mathbb{N}} A_m$.

We claim That $\bigcup_{m \in \mathbb{N}} A_m = \mathbb{N}$. The inclusion [$\subseteq$] follows for free because $\mathbb{N}$ is the universal set.
[$\supseteq$] Let $k \in \mathbb{N}$. Then $k = 1 \cdot k$ with $1 \in \mathbb{N}$, so $k \in A_1$.
Since $A_1 \subseteq \bigcup_{m \in \mathbb{N}} A_m$, we also have $k \in \bigcup_{m \in \mathbb{N}} A_m$.

   (c) Find $\bigcap_{m \in \mathbb{N}} A_m$.

We claim that $\bigcap_{m \in \mathbb{N}} A_m = \emptyset$. Assume for a contradiction That There is some $k \in \bigcap_{m \in \mathbb{N}} A_m$. Since $\mathbb{N}$ is the universal set, we have $k \in \mathbb{N}$. So also $k+1 \in \mathbb{N}$. Since $k \in \bigcap_{m \in \mathbb{N}} A_m \subseteq A_{k+1}$, we have $k \in A_{k+1}$. Thus, There exists some $q \in \mathbb{N}$ such That $k = (k+1)q$. Now we divide $k$ by $k+1$.
On one hand, $k = (k+1)q + 0$, so The remainder is 0.
On The other, $k = (k+1)0 + k$ with $0 \leq k < k+1$, so The remainder is $k$.
Since $k \neq 0$, This contradicts uniqueness of Div Thm.
Therefore, $\bigcap_{m \in \mathbb{N}} A_m = \emptyset$ □

5. (10 points) Let A and B be nonempty sets and $f: A \to B$ a function.

(a) Let $X \subseteq B$ and $Y \subseteq B$. Prove that $f^{-1}(X \setminus Y) = f^{-1}(X) \setminus f^{-1}(Y)$.

[$\subseteq$] Let $a \in f^{-1}(X \setminus Y)$. Then $f(a) \in X \setminus Y$. Hence $f(a) \in X$ and $f(a) \notin Y$, which say That $a \in f^{-1}(X)$ and $a \notin f^{-1}(Y)$, respectively. Hence $a \in f^{-1}(X) \setminus f^{-1}(Y)$.

[$\supseteq$] Let $a \in f^{-1}(X) \setminus f^{-1}(Y)$. Then $a \in f^{-1}(X)$ and $a \notin f^{-1}(Y)$. Hence $f(a) \in X$ but $f(a) \notin Y$. Thus $f(a) \in X \setminus Y$, which says That $a \in f^{-1}(X \setminus Y)$.

Typo! Should say $f(U \setminus V) \supseteq f(U) \setminus f(V)$

(b) Let $U \subseteq A$ and $V \subseteq A$. Prove that $f(U \setminus V) = f(U) \setminus f(V)$.

We show $f(U) \setminus f(V) \subseteq f(U \setminus V)$.

Let $b \in f(U) \setminus f(V)$. Since $b \in f(U)$, There is some $a \in U$ such That $f(a) = b$. Since $b \notin f(V)$ and $f(a) = b$, This $a$ cannot be in $V$.

Hence $a \in U \setminus V$. Therefore $b = f(a) \in f(U \setminus V)$.

Note: Inclusion $f(U \setminus V) \subseteq f(U) \setminus f(V)$ holds for injective $f$.

Conversely: if $f(U \setminus V) \subseteq f(U) \setminus f(V)$ for all $U, V \subseteq A$, Then $f$ needs To be injective.

6. (10 points) Let F be the set of all functions $f: \mathbb{R} \to \mathbb{R}$. Define a relation $\sim$ on F by $f \sim g$ if and only if there exists some $c \in \mathbb{R}$ such that $f(x) = g(x) + c$ for all $x \in \mathbb{R}$. Show that $\sim$ is an equivalence relation.

We need to show that $\sim$ is reflexive, symmetric and transitive

Reflexive: Let $f \in F$. Then, for the additive identity $c = 0 \in \mathbb{R}$, we have $f(x) = f(x) + 0$ $\forall x \in \mathbb{R}$. Hence $f \sim f$.

Symmetric: Let $f, g \in F$ so that $f \sim g$. Then there exists $c \in \mathbb{R}$ such that $f(x) = g(x) + c$ for all $x \in \mathbb{R}$. Then $g(x) = f(x) - c$ $\forall x \in \mathbb{R}$. Since $-c \in \mathbb{R}$, this means that $g \sim f$.

Transitive: Let $f, g, h \in F$ with $f \sim g$ and $g \sim h$. Let $c, d \in \mathbb{R}$ such that $f(x) = g(x) + c$ and $g(x) = h(x) + d$ for all $x \in \mathbb{R}$. Then, for any $x \in \mathbb{R}$ we have $f(x) = g(x) + c = h(x) + c + d$. Since $c + d \in \mathbb{R}$, this means that $f \sim h$.

Hence $\sim$ is an equivalence relation

Good extra practice:

1) What is the equivalence class of the function $f: \mathbb{R} \to \mathbb{R}$ such that $f(x) = 0$ $\forall x \in \mathbb{R}$?

2) What is the class of $g(x) = x$?

**Final Exam Notesheet**

**Divisibility.** For $m, n \in \mathbb{Z}$ with $m \neq 0$, $m$ *divides* $n$ if $n = mq$ for some $q \in \mathbb{Z}$. Denoted $m \mid n$.

(D1) Divisibility is reflexive and transitive.
(D2) If $a \mid b$ then $a \mid bc$.
(D3) The Division Theorem. For $a, b \in \mathbb{Z}$ with $b > 0$, there exists unique $q, r \in \mathbb{Z}$ with $a = bq + r$ and $0 \leqslant r < b$.

**Congruence.** For $n, a, b \in \mathbb{Z}$ with $n > 1$, we say $a \equiv b \pmod{n}$ if $n \mid (a - b)$.

(C1) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $a + c \equiv b + d \pmod{n}$.
(C2) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $ac \equiv bd \pmod{n}$.
(C3) If $a \equiv b \pmod{n}$ then $a^m \equiv b^m \pmod{n}$ for any $m \in \mathbb{N}$.
(C4) For any $a \in \mathbb{Z}$ there is a unique $r \in \mathbb{Z}$ with $0 \leqslant r < n$ such that $a \equiv r \pmod{n}$.

**Sets.** Let $A$, $B$ and $C$ be subsets of some universal set $U$.

(S1) $A \subseteq B$ when $x \in A \Rightarrow x \in B$.
(S2) $A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$.
(S3) $\mathcal{P}(A) = \{D \subseteq U \mid D \subseteq A\}$.
(S4) If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$ for any $n \geqslant 0$.
(S5) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
(S6) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
(S7) $(A^c)^c = A$ and $A \backslash B = A \cap B^c$.
(S8) $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$.

**Indexed Families of Sets.** Let $\Lambda$ be a not empty (index) set. Suppose that we have a set $A_\alpha$ for each $\alpha \in \Lambda$.

(IFS1) Union: $\bigcup_{\alpha\in\Lambda} A_\alpha = \{x \mid x \in A_\alpha \text{ for some } \alpha \in \Lambda\}$. Intersection: $\bigcap_{\alpha\in\Lambda} A_\alpha = \{x \mid x \in A_\alpha \text{ for every } \alpha \in \Lambda\}$.
(IFS3) For any $\beta \in \Lambda$, we have $A_\beta \subseteq \bigcup_{\alpha\in\Lambda} A_\alpha$ and $\bigcap_{\alpha\in\Lambda} A_\alpha \subseteq A_\beta$.
(IFS4) De Morgan's Laws: $\left(\bigcup_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcap_{\alpha\in\Lambda} A_\alpha^c$ and $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcup_{\alpha\in\Lambda} A_\alpha^c$.

**Functions.** Let $A$ and $B$ be non-empty sets.

(F1) A function $f\colon A \to B$ is injective if, for any $x, y \in A$, $f(x) = f(y) \Rightarrow x = y$.
(F2) A function $f\colon A \to B$ is surjective if for every $b$ in $B$ there exists some $a \in A$ such that $f(a) = b$.
(F3) A function is bijective if it is injective and surjective.
(F4) If $V \subseteq A$ we denote $f(V) = \{f(x) \mid x \in V\} \subseteq B$. If $Y \subseteq B$, we denote $f^{-1}(Y) = \{x \in A \mid f(x) \in Y\} \subseteq A$.
(F5) If $f\colon A \to B$ is a function we denote $f^{-1} = \{(f(a), a) \in B \times A \mid a \in A\}$.
(F6) If $f\colon A \to B$ is bijective then $f^{-1}$ defines a function $f^{-1}\colon B \to A$ defied by $f^{-1}(b) = a \Leftrightarrow f(a) = b$.
(F7) If $f\colon A \to B$ and $g\colon B \to C$, the composition $g \circ f\colon A \to C$ is defined by $(g \circ f)(x) = g(f(x))$.

**Relations.** Let $A$ and $B$ be sets. A relation $R$ from $A$ to $B$ is a subset $R \subseteq A \times B$. We denote $a \sim b$ to denote $(a, b) \in R$. A relation on a set $A$ is a relation from $A$ to $A$.

(R1) A relation $\sim$ on $A$ is an equivalence relation if it is:

- Reflexive: $x \sim x$ for all $x \in A$.
- Symmetric $x \sim y \Rightarrow y \sim x$.
- Transi: ($x \sim y$ and $y \sim z$) $\Rightarrow x \sim z$.

(R2) If $\sim$ is an equivalence relation, the equivalence class of $a \in A$ is $[a] = \{x \in A\colon x \sim a\}$.
(R3) If $\sim$ is an equivalence relation on $A$, the collection of equivalence classes is a partition of $A$.
(R4) For $n \in \mathbb{N}$, we denote by $\mathbb{Z}_n$ the set of equivalence classes for the relation of congruence modulo $n$.

**Equinumerousity.** Two sets $A$ and $B$ are equinumerous if there is a bijection $A \to B$. Denoted $A \approx B$.

(E1) Equinumerousity is an equivalence relation.
(E2) $A$ is finite if either $A = \varnothing$ or $A \approx \mathbb{N}_k = \{1, 2, \ldots, k\}$ for some $k \in \mathbb{N}$.
(E3) If $B$ is finite and $A \subseteq B$ then $A$ is finite and $\text{card}(A) \leqslant \text{card}(B)$.
(E4) If $A$ is equinumerous to one of its proper subsets, then $A$ is infinite.
(E5) Pigeonhole: If $A$ and $B$ are finite and $\text{card}(A) > \text{card}(B)$, there is no injection $A \to B$.
(E6) Surjective Pigeonhole: If $A$ and $B$ are finite and $\text{card}(A) < \text{card}(B)$, there is no surjection $A \to B$.
(E6) A set $A$ is countably infinite if $A \approx \mathbb{N}$. The sets $\mathbb{Z}$, $\mathbb{N} \times \mathbb{N}$ and $\mathbb{Q}$ are countably infinite.